THE U.S. NUCLEAR REGULATORY COMMISSION (NRC) is issuing this Sources Sought Synopsis as a means of conducting market research or as a market survey to determine the availability of potential qualified vendors with the capability to provide all technical, management, supervision, administration, and labor for subject matter expertise for an upcoming project titled "A Case Study for Using A Systems Engineering Approach to Demonstrate Assurance Against Common Cause Failure in Operating and New/Advanced Reactor Technologies."

The nuclear industry in the USA is adopting a Systems Engineering process, such as the Digital Engineering Guide (DEG) 3002011816, "Decision Making Using Systems Engineering," developed by the Electric Power Research Institute (EPRI). While the EPRI-developed guidance is intended to support industry practitioners and their management, there is a corresponding opportunity for the NRC to leverage a Systems Engineering approach in systematizing its migration towards risk-informed, performance-based regulatory reviews. This project is intended to develop part of the commensurate technical basis and framework in coordination with other NRC ongoing research efforts.

The objective of this work is to refine and validate a technical basis for assurance of protection against common cause failure (CCF) through a case study of a reactor trip system from the perspective of an independent evaluator or assessor. The technical basis (set of technical evaluation criteria) has been created in the projects A and B mentioned below. The criteria are performance-based (i.e., no prescriptive requirements such as diversity in design). The case study would evaluate safety requirements created in project A for performance-based safety assurance (e.g., no known hazardous conditions remain in the system). Examples of review considerations include hazardous interactions of the system with its environment, correct formulation of safety constraints, correct flow-down of the safety constraints, and search for any secondary contributing factors. The case study would also include voluntary reviews by external reviewers and consider their feedback on the safety of the design and the adequacy of the technical criteria.

The case study will build upon the results of three related projects:
A. Model-Based Systems Engineering (MBSE) research: The MBSE research develops technical criteria to evaluate the safety of a system developed using "correct-by-construction" (i.e., hazard-free) techniques. The research also includes an illustrative example satisfying the evaluation criteria.
B. Safety assurance case (SAC) research: The SAC research provides technical criteria to evaluate a SAC which integrates the evidence (e.g., as generated in the MBSE project) to demonstrate safety; its gap-identification capability will be useful in this project. Please see: https://sam.gov/opp/b6d774df34134e4b94246d4e0e6207d5/view
C. Hazard analysis (HA) research: The HA research identifies potential limitations in the capability to evaluate the hazard analysis performed by someone else using the System-Theoretic Process Analysis (STPA) method. Please see: https://sam.gov/opp/33ddd4e5fe8b437b9f90c2cc79a94915/view

For the uncontrolled hazards identified in the example design from project A, this case study will formulate the safety constraints to prevent those hazards, and rework the design or extend the verification as needed to assure its safety. Furthermore, the case study will be used to check whether any of the hazards identified remain uncontrolled due to gaps in the technical criteria. It would rework the technical criteria to fill any identified gaps.

This Sources Sought notice seeks information on the capability and availability of vendors experienced in providing research and development (R&D) services for designing and implementing systems engineering review processes in an organization that performs independent validation and verification (IV&V), safety assurance, certification, or licensing.

Services are to be provided remotely to the NRC located in Rockville, Maryland. The applicable North American Industry Classification System (NAICS) code assigned to this procurement is 541330, Engineering Services.

THERE IS NO SOLICITATION AT THIS TIME. This request for sources and vendor information does not constitute a request for proposal; submission of any information in response to this market survey is purely voluntary; the Government assumes no financial responsibility for any costs incurred.

REQUIRED CAPABILITIES:

1. Expertise in formulating and evaluating performance-based safety requirements sufficiently specific for engineering safety-critical cyber-physical systems.
2. Expertise in IV&V of high-criticality cyber-physical systems for the purpose of design certification or licensing in safety applications.
3. The offeror must have a minimum of 5 years of experience in performing or providing consulting support for activities associated with items #1, and #2, above, demonstrating:
    a. Ability to identify hazardous conditions introduced through the design, e.g., through interactions of the system with its environment (sensors; actuators; other systems and components; human-automation interactions) or through shared resources (signals; memory; computational resources; communication resources).
    b. Ability to formulate verifiable, unambiguous safety constraints to avoid or prevent hazardous conditions.
    c. Ability to evaluate whether architectural specifications satisfy safety constraints.
4. Offeror must be familiar with regulatory approval considerations for safety-critical systems in at least one application sector (e.g., aerospace, nuclear, transportation, medical, etc.).
5. Demonstrated capability for IV&V of safety-critical systems, especially for certification or licensing purposes.
6. Capability to perform a case study, which has used correct-by-construction model-based system and software engineering methods including V&V, to develop a safety-critical cyber-physical system, and organizing its safety analysis in the form of a safety assurance case.

7. A thorough understanding of NRC's research publications related to hazard analysis and safety assurance case for safety-critical systems such as reactor protection systems. Please see:
    a. RIL 1001, available at: https://www.nrc.gov/docs/ML1112/ML111240017.pdf
    b. RIL 1002, available at: https://www.nrc.gov/docs/ML1419/ML14197A201.pdf
    c. RIL 1101, available at: https://www.nrc.gov/docs/ML1423/ML14237A359.pdf
    d. NUREG/IA 0254, available at: https://www.nrc.gov/docs/ML1120/ML11201A179.pdf
    e. TLR-RES/DE/ICEEB-2025-002, available at: https://ww2.nrc.gov/docs/ML2524/ML25248A033.pdf

PREFERRED CAPABILITIES

In addition to the above required capabilities, the following capabilities would be preferred:

1. Expertise in transforming performance-based requirements into realized safety-critical cyber-physical systems that are correct-by-construction.
2. Demonstrated familiarity with the NRC's applicable regulations (e.g., 10 CFR 50.34).
3. The offeror should have a minimum of 5 years of experience in performing or providing consulting support for activities associated with required capability items #1, #2, and #3 above, demonstrating:
    a. Expertise in applying correct-by-construction techniques.
    b. Expertise in identifying hazards from systematic causes such as engineering deficiencies, including ability to utilize insights from NRC's research publications on this subject (mentioned above).
    c. Expertise in leveraging engineering models to identify hazards from engineering deficiencies and to formulate safety constraints that are verifiable analytically.

The purpose of this announcement is to provide potential sources the opportunity to submit information regarding their capabilities to perform work for the NRC free of conflict of interest (COI). For information on NRC COI regulations, visit NRC Acquisition Regulation Subpart 2009.5 (http://www.nrc.gov/about-nrc/contracting/48cfr-ch20.html).

All interested parties, including all categories of small businesses (small businesses, small disadvantaged businesses, 8(a) firms, women-owned small businesses, service-disabled veteran-owned small businesses, and HUBZone small businesses) are invited to submit a response. The capabilities package submitted by a vendor should demonstrate the firm's ability, capability, and responsibility to perform the principal components of work listed above. The package should also include past performance/experience regarding projects of similar scope listing the project title, general description, the dollar value of the contract, and name of the company, agency, or government entity for which the work was performed.

Organizations responding to this market survey should keep in mind that only focused and pertinent information is requested. If significant subcontracting or teaming is anticipated in order to deliver technical capability, organizations should address the administrative and management structure of such arrangements. Submission of additional materials such as glossy brochures or videos is discouraged.

HOW TO RESPOND TO THIS SOURCES SOUGHT NOTICE

If your organization has the capability and capacity to perform, as a prime contractor, one or more of the services described in this notice, then please respond to this notice and provide written responses to the following information. **Please do not include any proprietary or otherwise sensitive information in the response, and do not submit a proposal. Proposals submitted in response to this notice will not be considered.**

1. Organization name, address, emails address, Web site address and telephone number.

2. What size is your organization with respect to NAICS code identified in this notice (i.e., "small" or "other than small")? If your organization is a small business under the aforementioned NAICS code, what type of small business (i.e., small disadvantaged business, woman-owned small business, economically disadvantaged woman-owned small business, veteran-owned small business, service-disabled veteran-owned small business, 8(a), or HUBZone small business)? Specify all that apply.

3. Separately and distinctly describe how your organization meets each of the capabilities requirements specified in the above section, REQUIRED CAPABILITIES and PREFERRED CAPABILITIES. Please also provide the contract number, contract type, customer name, address and point of contact phone number and email address, contract value, thorough description of supplies and services included in the scope of that contract, indication of how they differ from the required capabilities described in this notice, period of performance (for services) your organization's role (prime contractor, first tier subcontractor, and/or supplier) in the related contracts, and any other relevant information.

4. Indicate whether your organization offers any of the required capabilities described in this notice on one or more of your company's own Federal Government contracts (i.e., GSA Federal Supply Schedule contract or Government wide Acquisition Contracts) that the NRC could order from and, if so, which services are offered. Also, provide the contract number(s) and indicate what is currently available for ordering from each of those contract(s).

5. Provide a standard, non-proprietary commercial price list or similar standard non-proprietary commercial pricing information for how your company sells, as a prime contractor, required capabilities described in this notice that your organization has experience providing. Also, indicate what is included in that pricing.

6. Is your organization currently performing or have in the past performed same or similar services as those listed above for any of the licensees regulated by the NRC? If so, which licensees? See http://www.nrc.gov/about-nrc/regulatory/licensing.html for more information on NRC licensing.

7. Has your organization previously faced organizational conflict of interest issues with NRC? If so, what were they and how were they mitigated or resolved?

Interested organizations responding to this Sources Sought Synopsis are encouraged to structure capability statements in the order of the area of consideration noted above. All capability statements sent in response to this notice shall not exceed 10 pages in length and must be submitted electronically, via e-mail, to Aracelis Pérez-Ortiz, at Aracelis.Perez-

Ortiz@nrc.gov, either MS Word or Adobe Portable Document Format (PDF), **by no later than February 26, 2026**. The subject line of the transmittal email must reference the notice ID number **APP-26-RES-0033**.

DISCLAIMER AND NOTES: **Any organization responding to this notice should ensure that its response is complete and sufficiently detailed to allow the Government to determine the organization's potential capability and capacity to perform the subject work. Respondents are advised that the Government is under no obligation to acknowledge receipt of the information received or provide feedback to respondents with respect to any information submitted. After a review of the responses received, a pre-solicitation synopsis and solicitation may be published in SAM.gov. However, responses to this notice will not be considered adequate responses to a solicitation.**